

NUMBER THEORY

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Number Theory

- Number theory: Study of the set of positive whole numbers. Eg 1, 2, 3, 4, Thus a set of Natural numbers N .
- Natural numbers can be classified into a variety of different types.
- Odd, Even, Perfect square, Cube, Prime, positive Integer, Congruent to $n \bmod m$, Fibonacci etc.
- Number Theory studies the different types of N , their relationships and proofs of these relationships.

- Numbers dividing 6 are: 1, 2, 3.

$$1 + 2 + 3 = 6.$$

- Numbers dividing 28 are: 1, 2, 4, 7, 14.

$$1 + 2 + 4 + 7 + 14 = 28.$$

Odd: congruent 1 mod 2.

Even: congruent 0 mod 2.

Prime: A prime number x is a number with two factors, 1 and x . (By Dirichlet)

Integer divisibility

- Integers: positive and negative whole numbers including a 0.
- If a and b are integers with $a \neq 0$, we say that a divides b if there is an integer c such that
- $b = ac$, or equivalently, if b/a is an integer.
When a divides b we say that a is a factor or divisor of b , and that b is a multiple of a . The notation $a \mid b$ denotes that a divides b . We write $a \nmid b$ when a does not divide b .

Integer divisibility theorem

- Let a , b , and c be integers, where $a \neq 0$. Then
- (i) if $a \mid b$ and $a \mid c$, then $a \mid (b + c)$;
- (ii) if $a \mid b$, then $a \mid b * c$ for all integers c ;
- (iii) if $a \mid b$ and $b \mid c$, then $a \mid c$.

Division Algorithm

- Let a be an integer and d a positive integer.
Then there are unique integers q and r , with $0 \leq r < d$, such that $a = dq + r$.
 d is called the divisor, a is called the dividend,
 q is called the quotient and r is called the remainder.
 $q = a \text{ div } d, r = a \text{ mod } d.$

- What are the quotient and remainder when 101 is divided by 11?
- $101 = 11 \cdot 9 + 2$.

Hence, the quotient when 101 is divided by 11 is

$9 = 101 \text{ div } 11$, and the remainder is

$2 = 101 \text{ mod } 11$.

Congruency Theorem

- If a and b are integers and m is a positive integer, then a is congruent to b modulo m if:
- m divides $a - b$.
- Notation $a \equiv b \pmod{m}$ to indicate that a is congruent to b mod m . We say that $a \equiv b \pmod{m}$ is a congruence.
- Let a and b be integers and let m be a positive integer. Then $a \equiv b \pmod{m}$ if and only if $a \bmod m = b \bmod m$.

- Determine whether 17 is congruent to 5 mod 6 and whether 24 and 14 are congruent mod 6.
- Because 6 divides $17 - 5 = 12$, we see that $17 \equiv 5 \pmod{6}$. However, because
- $24 - 14 = 10$ is not divisible by 6, we see that $24 \not\equiv 14 \pmod{6}$.
- Let m be a positive integer. The integers a and b are congruent mod m if and only if there is an integer k such that $a = b + km$.

Prime Factorization

Recall that an integer p is called a *prime* if:

- (i) $p \geq 2$.
- (ii) *The only positive divisors of p are 1 and p .*

The reason for not regarding 1 as a prime is that we want the factorization of every integer into primes to be unique. The following result is needed.

Prime Factorization

Theorem 2. 1. Euclid's Lemma. *Let p denote a prime.*

(i) If $p|mn$ where $m, n \in \mathbb{Z}$, then either $p|m$ or $p|n$.

(ii) If $p|m_1m_2 \cdots m_r$ where each $m_i \in \mathbb{Z}$, then $p|m_i$ for some i .

Proof. (i) Write $d = \gcd(m, p)$. Then $d|p$, so as p is a prime, either $d = p$ or $d = 1$.

If $d = p$, then $p|m$; if $d = 1$, then since $p|mn$, we have $p|n$ by Theorem 1.4 .

(ii) This follows from (i) using induction on r .

Prime Factorization

- **Theorem 2.2.** *Every integer $n > 1$ is a product of primes.*
- *Proof.* Let p_n denote the statement of the theorem. Then p_2 is clearly true.

If $p_2, p_3, \dots, p_k$ are all true, consider the integer $k + 1$. If $k + 1$ is a prime, there is nothing to prove. Otherwise, $k + 1 = ab$, where $2 \leq a, b \leq k$. But then each of a and b are products of primes because p_a and p_b are both true by the (strong) induction assumption. Hence $ab = k + 1$ is also a product of primes, as required.

Prime Factorization

- **Theorem 2.3. Prime Factorization Theorem.**
Every integer $n \geq 2$ can be written as a product of (one or more) primes. Moreover, this factorization is unique except for the order of the factors. That is,
if $n = p_1 p_2 \cdots p_r$ and $n = q_1 q_2 \cdots q_s$,
where the p_i and q_j are primes, then $r = s$ and the q_j can be relabeled so that $p_i = q_i$ for each i .

Prime Factorization

Collorary 2.4

If the prime factorization of n is $n = p_1^{n_1} p_2^{n_2} \cdots p_r^{n_r}$, then the positive divisors d of n are given as follows:

$$d = p_1^{d_1} p_2^{d_2} \cdots p_r^{d_r} \quad \text{where} \quad 0 \leq d_i \leq n_i \quad \text{for each } i.$$

Theorem 2.5 Suppose that m and n are positive integers, and write

$$\begin{aligned}n &= p_1^{n_1} p_2^{n_2} \cdots p_r^{n_r} & n_i &\geq 0 \\m &= p_1^{m_1} p_2^{m_2} \cdots p_r^{m_r} & m_i &\geq 0,\end{aligned}$$

where the p_i are distinct primes. Then:

$$\begin{aligned}\gcd(m, n) &= p_1^{\min(m_1, n_1)} p_2^{\min(m_2, n_2)} \cdots p_r^{\min(m_r, n_r)} \\ \operatorname{lcm}(m, n) &= p_1^{\max(m_1, n_1)} p_2^{\max(m_2, n_2)} \cdots p_r^{\max(m_r, n_r)}.\end{aligned}$$

The Euclidean algorithm works as follows. To find $g.c.d.(a, b)$, where $a > b$, we first divide b into a and write down the quotient q_1 and the remainder r_1 : $a = q_1b + r_1$. Next, we perform a second division with b playing the role of a and r_1 playing the role of b : $b = q_2r_1 + r_2$. Next, we divide r_1 into r_2 : $r_1 = q_3r_2 + r_3$. We continue in this way, each time

Proposition I.2.1. *The Euclidean algorithm always gives the greatest common divisor in a finite number of steps. In addition, for $a > b$*

$$\text{Time}(\text{finding } g.c.d.(a, b) \text{ by the Euclidean algorithm}) = O(\log^3(a)).$$

Example 1 (continued). To express 7 as a linear combination of 1547 and 560, we successively compute:

$$\begin{aligned}7 &= 28 - 1 \cdot 21 = 28 - 1(133 - 4 \cdot 28) \\&= 5 \cdot 28 - 1 \cdot 133 = 5(427 - 3 \cdot 133) - 1 \cdot 133 \\&= 5 \cdot 427 - 16 \cdot 133 = 5 \cdot 427 - 16(560 - 1 \cdot 427) \\&= 21 \cdot 427 - 16 \cdot 560 = 21(1547 - 2 \cdot 560) - 16 \cdot 560 \\&= 21 \cdot 1547 - 58 \cdot 560.\end{aligned}$$

Greatest Common Divisors

Let a and b be integers, not both zero.

The largest integer d such that $d \mid a$ and $d \mid b$ is called the **greatest common divisor** of a and b .

The greatest common divisor of a and b is denoted by $\gcd(a, b)$.

Example 1: What is $\gcd(48, 72)$?

The positive common divisors of 48 and 72 are
1, 2, 3, 4, 6, 8, 12, 16, and 24, so $\gcd(48, 72) = 24$.

Example 2: What is $\gcd(19, 72)$?

The only positive common divisor of 19 and 72 is
1, so $\gcd(19, 72) = 1$.

Using prime factorizations:

$a = p_1^{a_1} p_2^{a_2} \dots p_n^{a_n}$, $b = p_1^{b_1} p_2^{b_2} \dots p_n^{b_n}$,
where $p_1 < p_2 < \dots < p_n$ and $a_i, b_i \in \mathbf{N}$ for $1 \leq i \leq n$

$$\gcd(a, b) = p_1^{\min(a_1, b_1)} p_2^{\min(a_2, b_2)} \dots p_n^{\min(a_n, b_n)}$$

Example:

$$a = 60 = 2^2 3^1 5^1$$

$$b = 54 = 2^1 3^3 5^0$$

$$\gcd(a, b) = 2^1 3^1 5^0 = 6$$

PseudoRandom Numbers

Numbers generated to mimic random sequences
Used in simulations, randomized algorithms, etc.

Why "Pseudo"?

Generated by an algorithm
Not truly random—patterns may exist

Pseudorandom Numbers

Linear Congruential Generator:

$$x_{n+1} = (a \cdot x_n + c) \bmod m$$

a = multiplier

c = increment

x_0 = seed

m = modulus

Example:

$m = 9, a = 7, c = 4, x_0 = 3$

Generated sequence:

3, 7, 8, 6, 1, 2, 0, 4, 5, 3... (repeats)

Check Digits

Purpose:

Detect errors in identification numbers (books, tickets, products)

An extra digit is added for verification

Use congruences to calculate a "check digit"

If the digit doesn't match, there's an error

Check Digits

Example: Parity Check Bit

Used in digital transmission

Adds an extra bit to make the number of 1s even (or odd)

Helps detect single-bit errors

Other Uses:

Credit card numbers

Airline ticket numbers